Probability Bootcamp Summer 2024

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Contents

1 Lecture 1: Probability Measures and Random Variables **2**
1.1 Probability Triple $(\Omega, \mathcal{F}, \mathbb{P})$ . . . 2
1.2 Basic Properties of the Probability Measure . . . 3
1.3 Random Variables, Distribution Function, and Multivariate Extensions . . . 4
1.3.1 From Distribution Functions to Mass and Densities . . . 5

2 Lecture 2: Independence, Expectation and Moments **7**
2.1 Conditional Probabilities and Independence . . . 7
2.1.1 Probabilistic Events . . . 7
2.1.2 Extending to Random Variables . . . 8
2.2 Moments, Expectation and Variance . . . 9

1 Lecture 1: Probability Measures and Random Variables

1.1 Probability Triple $(\Omega, \mathcal{F}, \mathbb{P})$

When we model the *randomness* that occurs from physical experiments - meant in the broadest possible interpretation of the word - we with to quantify these notion of randomness that makes explicit our implicit common rational understanding of the notion of *probabilities*. Today, we review how we can use the mathematical frameworks from *set-theory* and *measure-theory* to make explicit these intuitions about randomness. In words,

probabilistic events take on a physical and geometric intuition, allowing that can be manipulated under a well-studied framework.

Let's make this precise by defining each constituent of the *probability triple* $(\Omega, \mathcal{F}, \mathbb{P})$, comprised of a sample space, σ-algebra, and probability measure respectively.

Definition 1.1: Sample Space: Ω

The set of all possible outcomes of an experiment is called the *sample space* and is denoted by Ω. We refer to specific outcomes as $\omega \in \Omega$.

Notice, this is a generic set without any structure. However, one is interested in being able to assign probabilities to events - that are combinations of outcomes in Ω. Morally, we need to allow operations such as

$$A \cup B, \quad A \cap B, \quad A^c,$$

where $A, B \subseteq \Omega$, as this captures the statements of A and B, A or B occurring, and not A occurring respectively. This motivates the following definition

Definition 1.2: σ-field: $\mathcal{F}$

A collection $\mathcal{F}$ of subsets of Ω is called a *σ-field* if it satisfies the following:

a) $\emptyset \in \mathcal{F}$;

b) If $A_1, A_2, \cdots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$;

c) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$.

If $A \subseteq \Omega$ is such that $A \in \mathcal{F}$, we say that A is a (measurable) *event*.

Warning. One is initially tempted to simply allow any subset of Ω, however, by *measure-theoretic considerations* we would **not be able to assign a probability measure to such a collection**. Specifically, it becomes possible to construct a pathological $A \subseteq \Omega$ such that $\mathbb{P}(A) = 0$ and $\mathbb{P}(A) = 1$.

Lastly, we consider a function that spits out the probability of any event in $\mathcal{F}$:

Definition 1.3: Probability Measure: $\mathbb{P}$

A *probability measure* $\mathcal{P}$ on $(\Omega, \mathcal{F})$ is a function $\mathbb{P} : \mathcal{F} \to [0, 1]$ satisfying

a) **Event of Everything and Nothing**: $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$;

b) **Sum of Constituent Subevents**: if $A_1, A_2, \cdots \in \mathcal{F}$ are pairwise disjoint, then

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

This codifies the intuition that mathematical probability allows us to *geometrize* events.

1.2 Basic Properties of the Probability Measure

One can easily derive the following properties from basic set-theory:

Lemma 1.1: Basic Properties of $\mathbb{P}$

The following hold:

a) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;

b) If $B \supseteq A$, then
$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A);$$

c) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;

Remark. This can be made more general.

Proof. Exercise. □

In fact, for the proof of (b) there is a general idea used. We will briefly discuss the general technique in the proof sketch of the following *technical* property of $\mathcal{P}$:

Lemma 1.2: Continuity of $\mathbb{P}$

Let $(A_i)_{i=1}^{\infty} \uparrow A$, i.e.,
$$A = \bigcup_{i=1}^{\infty} A_i = \lim_{i \to \infty} A_i,$$
then $\mathbb{P}(A) = \lim_{i \to \infty} \mathbb{P}(A_i)$.

Analogously, if $(B_i)_{i=1}^{\infty} \downarrow B$, i.e.
$$B = \bigcap_{i=1}^{\infty} B_i = \lim_{i \to \infty} B_i,$$
then $\mathbb{P}(B) = \lim_{i \to \infty} \mathbb{P}(B_i)$.

Proof. (*Sketch*) Disjointification. More specifically, we can see
$$A = A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus A_2) \cup \dots.$$
□

In these two lemmas, we are careful about overcounting/overlaps. However, it is often useful to have a coarse overestimate by not caring about overlaps, we record the following obvious but useful tool:

Lemma 1.3: Union Bound

Let $(A_i) \subseteq \mathcal{F}$ be a collection of events (finite or countably infinite). Then
$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

Moral Analysis Intuition. One generally considers taking approximations of some object of interest - in this setting this may probabilities, in other settings this may be functions. Analysis gives us many tools to bound this error, but one should be as cheap/coarse as the application allows. Some upper bounds can be extremely tight, but require a lot of assumptions to get working, when in reality one may not need to wrestle with such edge cases for a desired result to hold. In words, it is often better to be lazy and then tighten approximation bounds as needed, instead of the other way around.

1.3 Random Variables, Distribution Function, and Multivariate Extensions

Now that we have recalled how one models probabilities, we focus on the reason we even modeled these objects in the first place: *random variables.* Since we are not interested in the randomness of the outcomes themselves, but rather the consequence of those random outcomes,

random variables allow us to quantify randomness to these consequences of interest.

Morally, one can think of such random variables as *maps* from Ω to $\mathbb{R}$, which we denote $X : \Omega \to \mathbb{R}$, where we can assign probabilities numerical outcomes.

Naively, one would wish to assign a probability to every outcome $\omega \in \Omega$, that is one would like there to be an $f : \mathbb{R} \to [0, 1]$ such that

$$f(x) = \mathbb{P}(X = x).$$

However, this approach fails in general (try and remember why!). However, we can always attach with these outcomes a *distribution function* $F : \mathbb{R} \to \mathbb{R}$

$$F_X(x) = \mathbb{P}\{\omega \in \Omega : X(\omega) \leq x\}.$$

Since $\mathbb{P}$ is a function on $\mathcal{F}$, this motivates the following definition for a *random variable*

Definition 1.4: Random Variable

A *random variable* is a function $X : \Omega \to \mathbb{R}$ such that

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

for all $x \in \mathbb{R}$. Such a function is said to be $\mathcal{F}$-measurable.

Remark. One can show more rigorously that such a definition allows us to assign probabilities to consequences of events as desired. Namely, we are able to make an association between $\mathcal{F}$ and $\mathcal{B}$, the Borel σ-algebra on $\mathbb{R}$.

We will return to the focus on random variables in the next lecture, before then we will focus on the motivating object of the definition for the random variable, the *distribution function*:

Definition 1.5: Distribution Functions

The *distribution function* of a random variable X is the function $F : \mathbb{R} \to [0, 1]$ given by $F_X(x) = \mathbb{P}(X \leq x) := \mathbb{P}\{\omega \in \Omega : X(\omega) \leq x\}$.

These are incredibly important associated functions to random variables. As such, we compile some characterizations and consequences of the definition.

Lemma 1.4: Equivalent Characterization of Distribution Function

A function $F : \mathbb{R} \to [0, 1]$ is a *distribution function* if and only if

1. $\lim_{x\to-\infty} F(x) = 0$, and $\lim_{x\to+\infty} F(x) = 1$;
2. If $x < y$, then $F(x) \leq F(y)$;
3. F is right-continuous, that is $\lim_{h\downarrow 0} F(x + h) = F(x)$

Remark. This captures the usual visual intuition of a distribution function.

The following makes explicitly clear the remark from earlier, the distribution function really connects the outputs of random variable X with the associated probability triple $(\Omega, \mathcal{F}, \mathbb{P})$.

Lemma 1.5: Consequences of Definition

Let F be a distribution function of X. Then

a) $\mathbb{P}(X > x) = 1 - F(x)$;

b) $\mathbb{P}(x < X \leq y) = F(y) - F(x)$;

c) $\mathbb{P}(X = x) = F(x) - \lim_{y \uparrow x} F(y)$.

From these consequences, we recover how the probabilities *tail events* of our random variable - that is the extreme values - can be quantified through the distribution function. In modern applications, we are quite interested in recovering quantitative bounds on different types of events. We will recall some basic non-asymptotic bounds on concentration in a future lecture, as well as hint at what is used at the forefront of modern research. We now highlight how we can these distribution functions also allow us to recover probabilities for the outcomes of collections of random variables - through *random vectors*.

Consider how a random variable X has an associated distribution function F_X defined by $F_X(x) = \mathbb{P}(X \leq x)$ for all $x \in \mathbb{R}$. Analogously, suppose we have a random vector $(X_1, X_2, \ldots, X_n) : \Omega \to \mathbb{R}^n$, where each component is in it's own right is a random variable. Then, $(X_1, \ldots, X_n)$ is a *random vector* with the corresponding *joint* distribution function

$$\mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \ldots, X_n \leq x_n),$$

where $x_1, \ldots, x_n \in \mathbb{R}$ are variables.

Remark. There is a nice geometric interpretation of the above distribution function if the components are *independent* of one another. More on this later.

Using boldface $\mathbf{X}$ and $\mathbf{x}$ to refer to the vector-valued versions of random variables and variables, and $\mathbf{x} \leq \mathbf{y}$ to denote $x_i \leq y_i$ for all $i \in [n]$, we have the following:

Definition 1.6: Joint Distribution Function

The *joint distribution function* of a random vector $\mathbf{X} = (X_1, \ldots X_n)$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is the function $F_{\mathbf{X}} : \mathbb{R}^n \to [0, 1]$ given by $F_{\mathbf{X}}(x) = \mathbb{P}(\mathbf{X} \leq \mathbf{x}) = \mathbb{P}(\{\omega \in \Omega : \mathbf{X}(\omega) \leq \mathbf{x}\})$ for all $\mathbf{x} \in \mathbb{R}^n$.

One can easily define analogous consequences of Lemma 1.4 (guess!), as the construction did not depend on the dimensionality of the random vector to be 1.

1.3.1 From Distribution Functions to Mass and Densities

We finish with recalling the connections with the distribution function with the familiar concepts heavily emphasized in an undergraduate probability course: *probability mass functions* (discrete probabilities and discrete random variables) and *probability densities* (continuous probabilities and continuous random variables). Before recalling these notions, we make clear

the distinction between discrete vs. continuous is (mostly) negligible - when considering the framework of measure theory.

While there are strong distinctions between discrete and continuous, when we are considering the *framework* of probability theory, measure theory gives us a way to ***compress*** the main ideas and tools being used. The distinction manifests in the computations carried out - and thus specific outcomes and inferences from said computations - but morally the logic behind the tools used are unified by the notation of framework of measure theory.

Remark. In undergraduate probability, one goes through the exact same constructions twice, once for discrete probabilities/random variables and again for the continuous analog. It should not be surprising that since the logic is precisely the same that the differences are somewhat negligible once sufficiently abstracted.

To make this point clear, suppose we wish to quantify the probability of a *specific outcome*, or in other words an infinitesimal consequence.

Definition 1.7: Discrete Random Variables and Probability Mass Functions

The random variable X is called *discrete* if it takes on the values in some at most ***countable*** subset of values of $\mathbb{R}$, denoted $\{x_i\}_{i=1}^{\infty}$. Then, the discrete random variable X has a a *probability mass function* $f : \mathbb{R} \to [0, 1]$ given by

$$f_X(x) = \mathbb{P}(X = x).$$

Definition 1.8: Continuous Random Variables and Probability Density Functions

The random variable X is called *continuous* if its distribution function can be expressed as

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f_X(u)du, \quad x \in \mathbb{R},$$

for some $f : \mathbb{R} \to [0, \infty)$, an integrable function called the *probability density function of* X.

Notice, that this interpretation also holds for the discrete random variables, where the integral becomes a summation: namely in the discrete setting

$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{x_i : x_i \leq x} \mathbb{P}(X = x_i).$$

In fact, one may have random variables that are a mixture of discrete and continuous random variables. We note that these definitions are **referring to properties of distribution functions, rather than the random variables (the functions) themselves**.

The extensions to random vectors is also immediate, we merely state the analogs to the probability mass functions and density functions, respectively:

$$f_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(X_1 = x_1, \ldots, X_n = x_n),$$

and in the continuous case $f \in L^1_{\text{ac}}(\mathbb{R}^n)$ such that

$$F_{\mathbf{X}}(\mathbf{x}) = \mathbb{P}(\mathbf{X} \leq \mathbf{x}) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(x_1, \ldots, x_n) dx_1 \ldots dx_n.$$

In more generality, one can unify all of these ideas with the following notation:

$$\mathbb{P}(\mathbf{X} \leq \mathbf{x}) = \mathbb{P}(\{\omega \in \Omega : \mathbf{X}(\omega) \leq \mathbf{x}\}) = \int_{\omega : X(\omega) \leq \mathbf{x}} d\mathbb{P}(\omega) = \int_{\Omega} 1_{\omega : X(\omega) \leq \mathbf{x}} d\mathbb{P}(\omega).$$

where the notation $d\mathbb{P}(\omega)$ refers to the probability mass/density that is assigned to the infinitessimal outcome denoted by ω. The subscript on the integral denotes what event we are considering, and thus we can interpret the integral as *adding up the probabilities of all of the outcomes* that fall under the event defined by the random variable/vector - or in words the consequence in question. We will return to these more general integrals in the next lecture.

2 Lecture 2: Independence, Expectation and Moments

In this lecture, we focus on two ideas, what is independence for events and random variables, and how can we make a summary of the behavior of a random variable?

2.1 Conditional Probabilities and Independence

2.1.1 Probabilistic Events

First, let's recall the notion of a *conditional probability* for a first pass - to codify how the probability of an event depends on another:

Definition 2.1: Conditional Probability

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple. Let $A, B \in \mathcal{F}$ such that $\mathbb{P}(B) > 0$. Then, the *conditional probability* that A occurs given that B occurs is defined as

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

In a similar fashion to the *disjointification idea* of before, using simple ideas from *set-theory* we can consider a partition of Ω. Let $\{B_i\}_{i=1}^n$ be a partition of Ω such that for all $i \neq j$,

$$B_i \cap B_j = \emptyset \quad \text{and} \quad \bigcup_{i=1}^n B_i = \Omega.$$

This allows us to *bin* every $\omega \in \Omega$ into exactly one set of this partition. Thus,

Lemma 2.1: Law of Total Probability

For any events A and B such that $0 < \mathbb{P}(B) < 1$,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c).$$

More generally, let $B_1, \ldots, B_n$ be some partition of Ω such that $\mathbb{P}(B_i) > 0$ for all i. Then

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Proof. (*Scheme*) Use a disjoint union and definition of conditional probability. □

Definition 2.2: Independent Events

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple. Then $A, B \in \mathcal{F}$ are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

More generally, a family $\{A_i : i \in I\}$ is called *independent* if

$$\mathbb{P}\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \mathbb{P}(A_i)$$

for all finite subsets J of I.

Remark. This does not imply that $A \cap B = \emptyset$.

Morally, this captures the notion of the outcome of an event as not being dependent on the other event. Moreover, we may even weaken the notion of independent events to *pairwise independence*

Definition 2.3: Pairwise Independence

Let $\{A_i\}_{i \in I} \subseteq \mathcal{F}$ such that for all $i \neq j$ we have

$$\mathbb{P}(A_i \cap A_j) = \mathbb{P}(A_i)\mathbb{P}(A_j).$$

Then, we say that this family is *pairwise independent.*

These notions become incredible useful when trying to understand complicated probabilistic events. Recall, last lecture we highlighted the general structure of probability spaces can be understood through the lens of *set theory* and *measure theory.* Some arguments required decomposition arguments, or in words breaking down an event into manageable and reasonable pieces. Conditional probabilities also play a role on understanding relationships between events, and independence highlights a scenario where it is very easy to manipulate the randomness. This will pop up when talking about some fundamental properties of expectation and variance, more on this later.

2.1.2 Extending to Random Variables

There are many notions of independence of random variables that are taught in classes. However, recall in the first lecture recall that we were able to explicitly connect the outputs of our random variables with their sources of randomness through the *distribution function.* In light of the previous discussion, the following definition is clear

Definition 2.4: Independent Random Variables

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability triple. Let $X, Y : \Omega \to \mathbb{R}$ be random variables with distribution function $F_X : \mathbb{R} \to [0,1]$ and $F_Y : \mathbb{R} \to [0,1]$. Consider the joint distribution function $F_{X,Y}$ on $(X,Y) : \mathbb{R} \times \mathbb{R} \to [0,1]$. We say X, Y are *independent* if and only if

$$F_{X,Y}(x,y) = F_X(x)F_Y(y) \quad \forall x, y \in \mathbb{R}.$$

Remark. We can read this as $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, where we define the events A and B accordingly.

Recall, that it suffices to consider the closed rays in $\mathbb{R}$ as this generates the Borel σ-algebra, and the distribution function is a *nice* function that allows us to transfer our understanding from $\mathcal{B}$ to $\mathcal{F}$. In words, we recover *independence* in the sense of the last discussion for our random variables - and we use the fact the distribution function carries the information of the randomness of our random variables for this to work.

For completeness, we list the other familiar notions of independence for random variables in the discrete and continuous case, stating that these are special cases of the above definition (why?):

- **Discrete Random Variables**: X and Y are independent if and only if events

  $$\{X = x\} \quad \text{and} \quad \{Y = y\}$$

  are independent for all x, y;

- **Continuous Random Variables**: X and Y are independent if and only if events

  $$\{X \leq x\} \quad \text{and} \quad \{Y = y\}$$

  are independent for all x, y.

We also note that one can recover the *factorization* of not just the joint distribution functions, but also the joint probability mass functions and joint probability densities.

Remark. This factorization hints at an underlying geometric interpretation, where independence is precisely the setting of being able to take the *tensor product* of some spaces. In general, the *product measure* marginalizes well, and it is precisely this product measure that reflects the notion of independence.

The nice outcomes of independence, and thus factorization, can truly be felt when considering outcomes in expectation. Beforehand, we make note of a notion that will be useful when discussing limiting phenomenon - such as the *Central Limit Theorem*

Definition 2.5: Independently Identically Distributed (IID) Random Variables

Let $\{X_i\}_{i \in I}$ be a collection of random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Then, we say that this collection is *independently identically distributed* if they are *independent* of a collection of random variables, and have the *same distribution function*.

This definition is quite natural, if we assume we are making independent samples of some quantity governed by the same underlying distribution.

Remark. In modern regimes, we wish to relax this situation - for example in *online learning* - but we can still stay close to this case in some suitable sense of *close*.

2.2 Moments, Expectation and Variance

One often wishes to have a *single scalar* descriptions of the distribution of values taken by some random variable $X : \Omega \to \mathbb{R}$. To produce such scalar descriptions to describe notions such as the *center of mass*, *spread of mass* - given that we conceptualize the outputs of the random variable as mass - we use the *moments of a random variable*

Definition 2.6: Moments of a Random Variable

Let $X : \Omega \to \mathbb{R}$ be a random variable, and $n \in \mathbb{N}$. The *p-th moment* of X is defined as

$$\mathbb{E}[X^p] := \int_\Omega (X(\omega))^p d\mathbb{P}(\omega).$$

As hinted by the notation, the *expectation* of X is the 1st moment of X. Explicitly:

$$\mathbb{E}[X] := \int_\Omega X(\omega) d\mathbb{P}(\omega).$$

A few remarks. First, we use the notation introduced at the end of the previous section alluding to the use of the *Lebesgue integral*. Morally, one can still have the same intuitions of integration from Calculus, but this allows one to integrate more nasty functions - one can learn a more rigorous representation in the Probability Theory sequence and/or a Graduate Analysis course. Second, it is intuitive that the moments we have, the more specified our distribution is. This idea plays a role in estimating distributions from empirical samples. Moreover, it is important to state that in modern applied probability, the notion of *Gaussian Universality* essentially says that having the first two moments may suffice in practice. These ideas are still being fleshed out in the community, with many open problems!

Notice that the expressions for the expectation of discrete and continuous random variables are just special cases, respectively:

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}) \quad \text{and} \quad \mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx,$$

where we explicitly use the *probability mass function* and *probability density function*.

Before going through the standard properties, we mention some nice identities that can be used often:

- **Expectation of an Indicator.** Let $A \in \mathcal{F}$ be arbitrary. Then, if we define the *indcator function* I_A as
$$I_A(\omega) := \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A, \end{cases}$$
then we have $\mathbb{E}[I_A] = \mathbb{P}(A)$. Explicitly, notice
$$\mathbb{E}[I_A] = \int_\Omega I_A(\omega) d\mathbb{P}(\omega) = \int_A d\mathbb{P}(\Omega) = \mathbb{P}(A).$$
In other words, we are able to convert the probability of an event into an expectation - giving a dual perspective on how to manipulate these objects;

  Remark. In fact, one can arbitrarily approximate any random variable through the use of linear combinations of indicator functions in a dense fashion - take Probability Theory to see such an idea.

- **Tail Identity.** Let X be a *non-negative* random variable. Then,
$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > x) dx.$$
To prove this, one uses *Fubini-Tonelli's Theorem*
$$\int_0^\infty \mathbb{P}(X > x) dx = \int_0^\infty \int_\Omega I_{\{X>x\}}(\omega) d\mathbb{P}(\omega) = \int_\Omega \int_0^\infty I_{\{X>x\}} dx d\mathbb{P}(\omega) = \int_\Omega X(\omega) d\mathbb{P}(\omega) = \mathbb{E}[X].$$
The only part that needs to be justified is switching the order of integration. Note, in the simple continuous or discrete case this reduces to the usual computations.

  We note that this identity generalizes even further, without proof. If X is any random variable, and Φ is any increasing and differentiable function Φ, then we have
$$\mathbb{E}[\Phi(|X|)] = \Phi(0) + \int_0^\infty \Phi'(t) \mathbb{P}[|X| > t] dt.$$
This is useful for proving properties for *subgaussian random variables*, which are an important class of random variables in Statistics and Machine Learning.

  Remark. The proof follows essentially from an application of Fundamental Theorem of Calculus.

We now recall familiar properties that follow directly from nice properties of the integral.

Lemma 2.2: Basic Properties of Expectation

a) **Linearity:** Let X, Y be random variables and $\alpha, \beta \in \mathbb{R}$, then we have
$$\mathbb{E}[\alpha X + \beta Y] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y];$$

b) **Deterministic Random Variables:** Suppose that $\mathbb{P}(X = b) = 1$ for some $b \in \mathbb{R}$, then one has
$$\mathbb{E}[X] = b.$$

c) **Bounded Random Variables:** Suppose that $\mathbb{P}(a \leq X \leq b) = 1$ for some $a, b \in \mathbb{R}$, then one has
$$a \leq \mathbb{E}[X] \leq b.$$

Proof. (*Scheme*) Follows from the definition of expectation. □

In addition, suppose we have some measurable function $g : \mathbb{R} \to \mathbb{R}$ - measurable in our context means some reasonable function - then we can easily compute the expectation of $g(X)$ when X is a random variable:

Lemma 2.3

Suppose that $g : \mathbb{R} \to \mathbb{R}$ is measurable, and X is a random variable. Then, we have the following realizations of the expectation: First, if X is discrete then

$$\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x).$$

If continuous with probability density function f_X, then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

More generally, we can write this as

$$\mathbb{E}[g(X)] = \int_\Omega g(X(\omega)) d\mathbb{P}(\omega) = \int_{-\infty}^{\infty} g(x) dF_X(x),$$

where in the last expression we use the distribution function $F_X(x)$ of random variable X.